

Tutorial 11

Matrix representations and eigenspaces

1. For each of the following linear transformations $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ find a 2×2 matrix such that for each $\mathbf{x} \in \mathbb{R}^2$, $T(\mathbf{x}) = A\mathbf{x}$.

(a) $T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} -x_1 \\ x_2 \end{pmatrix}$

Solution: $A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

(b) $T\left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right) = \begin{pmatrix} x_2 \\ -x_1 \end{pmatrix}$

Solution: $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$

2. In this question we go back to two questions in Tutorial 10. You may have to look at your work for those questions to answer this one.

- (a) Find the matrix representing the linear transformation T in Question 2 of Tutorial 10 with respect to the basis B given in that question for $P_2(x)$ and the standard basis $C = \{1, x\}$ for $P_1(x)$.

Solution: It should be the matrix $\begin{pmatrix} 2 & 0 & 4 \\ -1 & 3 & 1 \end{pmatrix}$.

- (b) Find the matrix ${}_C[T]_B$ representing the linear transformation T in Question 3 of Tutorial 10 with respect to the standard basis B for $M_{2 \times 2}(\mathbb{R})$ and the standard basis C for P_2 .

Solution: By Theorem 3.3.5, the columns of ${}_C[T]_B$ are the coordinate vectors of the polynomials x^2 , $x^2 + x$, $x + 1$, $-x - 1$ (see the calculations in the solutions of Tutorial 10, Question 3(f)) with respect to the standard basis $\{1, x, x^2\}$ for P_2 . Hence

$${}_C[T]_B = \begin{pmatrix} 0 & 0 & 1 & -1 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

3. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 2 \\ -2 & 0 & 3 \end{pmatrix}$.

(a) Find all the eigenvalues of A .

Solution: The characteristic equation for A is

$$\det(\lambda I - A) = (\lambda - 1)^2(\lambda - 3) = 0.$$

Hence the eigenvalues of A are $\lambda = 1, 3$.

(b) For each eigenvalue λ found in (a), find the nullity of $A - \lambda I_3$.

Solution: For $\lambda = 1$ we have

$$A - \lambda I_3 = A - I_3 = \begin{pmatrix} 0 & 0 & 0 \\ -2 & 0 & 2 \\ -2 & 0 & 2 \end{pmatrix} \text{ with row-echelon form } U = \begin{pmatrix} -2 & 0 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence the nullity of $A - I_3$ is 2 (the number of free variables in $U\mathbf{x} = \mathbf{0}$).

For $\lambda = 3$ we have

$$A - \lambda I_3 = A - 3I_3 = \begin{pmatrix} -2 & 0 & 0 \\ -2 & -2 & 2 \\ -2 & 0 & 0 \end{pmatrix} \text{ with row-echelon form } U = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Hence the nullity of $A - 3I_3$ is 1.

(c) For each eigenvalue λ found in (a), find a basis for $NS(A - \lambda I_3)$.

Solution: $NS(A - I_3) = \left\{ s \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} : s, t \in \mathbb{R} \right\}$ with basis $\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$

and

$$NS(A - 3I_3) = \left\{ t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} : t \in \mathbb{R} \right\} \text{ with basis } \left\{ \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

(d) Are the basis vectors you found in (c) (there should be 3 in total) linearly independent? Give a reason for your answer.

Solution: Let $P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$ be the matrix with the three basis vectors

as columns. Then $\det(P) = 1 \neq 0$, and so the basis vectors are linearly independent.

- (e) Is the 3×3 matrix P with these basis vectors as columns invertible? Give a reason for your answer. If it is invertible, find $P^{-1}AP$.

Solution: Since $\det(P) \neq 0$, P is invertible. Check that

$$P^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & -1 \\ -1 & 0 & 1 \end{pmatrix} \text{ and } P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

4. Let $0 < \theta < 2\pi$ be fixed and A_θ be the 2×2 matrix such that for every $\mathbf{x} \in \mathbb{R}^2$, $A_\theta \mathbf{x}$ is the vector obtained by rotating $\mathbf{x}$ anti-clockwise about the origin through θ radians.

- (a) Show that there is only one value of θ for which A_θ has real eigenvalues. (You should be able to do this without doing any calculations.)

Solution: For $\theta = \pi$, -1 is an eigenvalue, since $A\mathbf{x} = -1\mathbf{x}$ for any non-zero $\mathbf{x}$. For all other values of θ , rotating a non-zero vector $\mathbf{x}$ through θ radians will not give a scalar multiple of $\mathbf{x}$.

- (b) Show that for all other values of θ A_θ has complex eigenvalues, and find them.

Solution: We have $A_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ and

$$0 = \det(\lambda I_2 - A_\theta) = (\lambda - \cos \theta)^2 + \sin^2 \theta = \lambda^2 - 2\lambda \cos \theta + 1,$$

and hence the eigenvalues are $\lambda = \cos \theta + i \sin \theta$ or $\lambda = \cos \theta - i \sin \theta$.

5. Let $A = \begin{pmatrix} 2 & 0 & 0 \\ -2 & 4 & 2 \\ 2 & -1 & 1 \end{pmatrix}$.

(a) Find all the eigenvalues of A .

Solution: To calculate the eigenvalues of A , we solve:

$$0 = \begin{vmatrix} \lambda - 2 & 0 & 0 \\ 2 & \lambda - 4 & -2 \\ -2 & 1 & \lambda - 1 \end{vmatrix} = (\lambda - 2)[(\lambda - 4)(\lambda - 1) + 2] = (\lambda - 2)(\lambda - 2)(\lambda - 3),$$

hence the eigenvalues are $\lambda = 2, \lambda = 3$.

(b) For each eigenvalue λ of A , find the corresponding eigenspace

$$W_\lambda = NS(\lambda I_3 - A)$$

and a basis for it.

Solution: For $\lambda = 2$ we have $W_2 = NS(2I - A)$ and $2I - A = \begin{pmatrix} 0 & 0 & 0 \\ 2 & -2 & -2 \\ -2 & 1 & 1 \end{pmatrix}$

reduces to $\begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$. Hence $W_2 = \left\{ t \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} : t \in \mathbb{R} \right\}$ and a basis for W_2 is $\left\{ \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} \right\}$.

For $\lambda = 3$ we have $W_3 = NS(3I - A)$ and $3I - A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -1 & -2 \\ -2 & 1 & 2 \end{pmatrix}$ reduces to $\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{pmatrix}$. Hence $W_3 = \left\{ t \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} : t \in \mathbb{R} \right\}$ and a basis for W_3 is $\left\{ \begin{pmatrix} 0 \\ -2 \\ 1 \end{pmatrix} \right\}$.

(c) How many linearly independent eigenvectors does A have?

Solution: A has only 2 linearly independent eigenvectors.

6. Let $B = \begin{pmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ -4 & -1 & -8 \end{pmatrix}$.

- (a) You are told that 9 and -9 are eigenvalues of B . Find the other eigenvalue of B *without* using the characteristic equation of B .

Solution: The sum of the eigenvalues of B equals the trace of B . Since $\text{trace}(B) = -9$ and we are given that 9 and -9 are eigenvalues, the third eigenvalue must be -9 .

- (b) For each eigenvalue λ of B , find the eigenspace $W_\lambda = NS(\lambda I_3 - B)$ and a basis for it.

Solution: $W_9 = \left\{ \begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix} t : t \in \mathbb{R} \right\}$ and a basis for W_9 is $\left\{ \begin{pmatrix} 4 \\ 1 \\ -1 \end{pmatrix} \right\}$.
 $W_{-9} = \left\{ s \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} + t \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} : s, t \in \mathbb{R} \right\}$, and a basis for W_{-9} is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix}, \begin{pmatrix} -1 \\ 4 \\ 0 \end{pmatrix} \right\}$.

- (c) Does $\mathbb{R}^3$ have a basis consisting of eigenvectors of B ? Give a reason for your answer.

Solution: Yes. B has 3 linearly independent eigenvectors, and since $\dim(\mathbb{R}^3) = 3$, any 3 linearly independent vectors will span $\mathbb{R}^3$.